



Assignment 3 , matlab and simulink

Linear Control Systems

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Part (c): MATLAB Verification Using Step Responses

In this part, the objective is to verify the stability analysis obtained in the previous sections through simulation. For each of the six given systems, the step response was generated using MATLAB. All responses were plotted together in a single figure in order to directly compare their dynamic behaviors.

The step response provides a time-domain visualization of how each system reacts to a bounded input. By observing whether the outputs remain bounded, grow without bound, or exhibit sustained oscillations, the simulation serves as a practical confirmation of the theoretical stability results previously derived using the impulse response norm criterion.

The combined plot of the six step responses demonstrates behavior that is fully consistent with the expected stability properties of the systems, thereby validating the analytical results through numerical simulation.

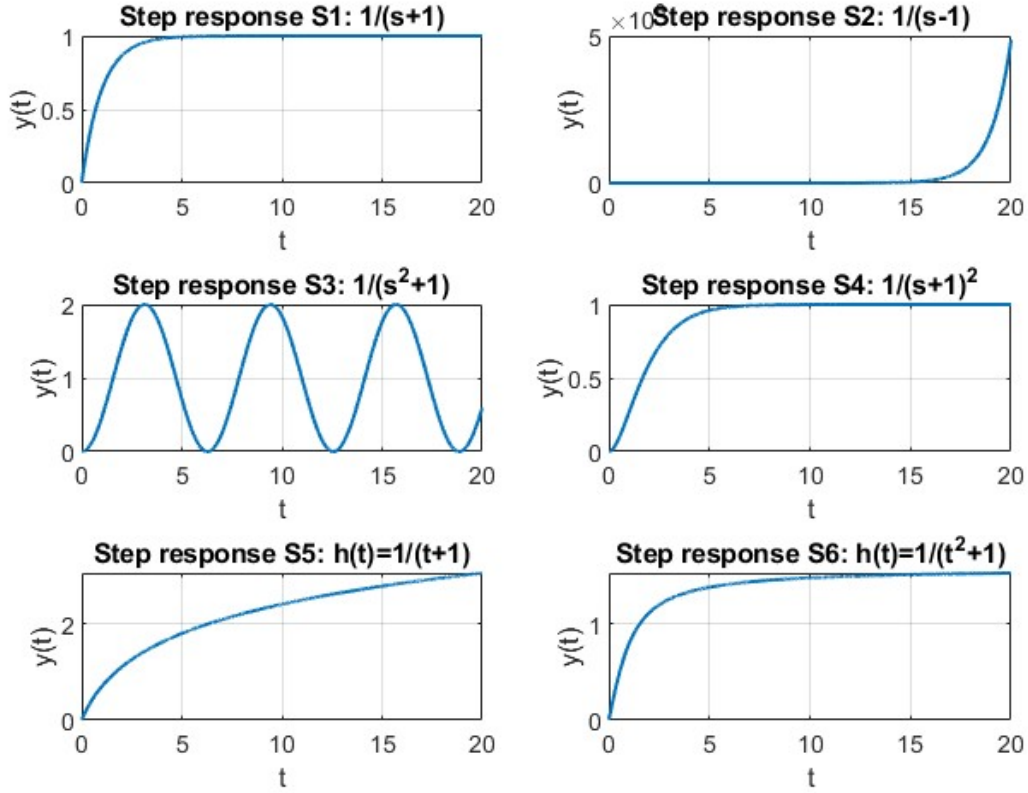


Figure 1: Overlay comparison of all identified first-order transfer functions $G_i(s) = \frac{k_i}{s+a_i}$.

First-Order System Identification and Validation

The goal of this study is to approximate the system dynamics by a first-order continuous-time transfer function of the form

$$G(s) = \frac{k}{s+a}, \quad (1)$$

where k denotes the static gain and $a > 0$ determines the pole location and dominant time constant $\tau = 1/a$.

(a) Measured dataset preparation (Data1)

The measured input-output data were imported from a MAT-file and converted into an iddata object. A robust loading procedure was employed to automatically detect the input, output, and sampling time variables, allowing the script to function correctly even when variable names differ. The resulting dataset, denoted as Data1, serves as the reference dataset for system identification.

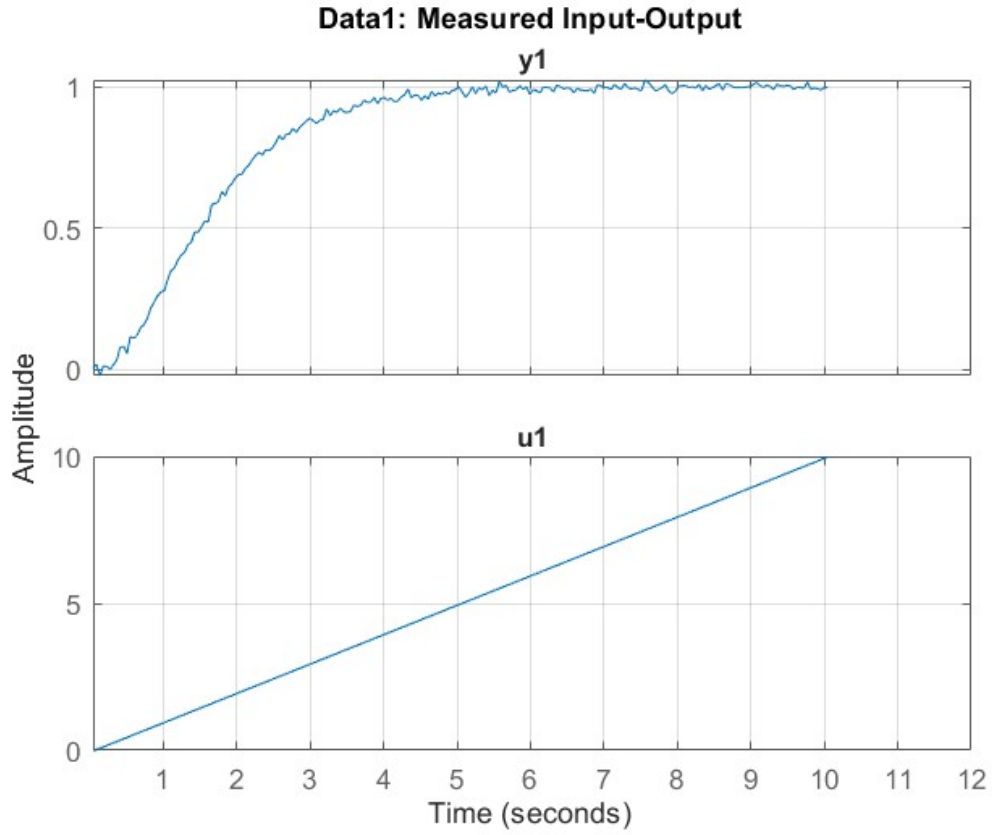


Figure 2: Measured system response corresponding to Data1.

(b) Transfer function estimation from measured data

Using the System Identification Toolbox, a first-order transfer function model with one pole and zero zeros was estimated from Data1 using the `tfest` routine. This yields the model

$$\hat{G}_1(s) = \frac{\hat{k}_1}{s + \hat{a}_1}, \quad (2)$$

where the parameters $\hat{k}_1$ and $\hat{a}_1$ are obtained directly from the estimated numerator and denominator coefficients.

```

Command Window
>> tfest ( data1 , 1 , 0 )

ans =

    From input "u1" to output "y1":
    60.49
    -----
    s + 419.5

Continuous-time identified transfer function.

Parameterization:
    Number of poles: 1    Number of zeros: 0
    Number of free coefficients: 2
    Use "tfdata", "getpvec", "getcov" for parameters and their uncertainties.

Status:
    Estimated using TFEST on time domain data "Data1 (measured)".
    Fit to estimation data: -1.012%
    FPE: 0.0858, MSE: 0.08347

Model Properties
fx >> | Search - Day of the Poinsettia

```

Figure 3: First-order transfer function estimated from Data1 using tfest.

(c) Sine excitation dataset (Data2)

To investigate the system behavior under a structured excitation, the original input signal was replaced by a sinusoidal input while preserving the original sampling time and data length. The corresponding output response forms the dataset Data2. This dataset represents the system response to a single-frequency excitation and is used to evaluate consistency of the identified dynamics.

A first-order transfer function was estimated from Data2 using the same identification structure, resulting in

$$\hat{G}_2(s) = \frac{\hat{k}_2}{s + \hat{a}_2}. \quad (3)$$

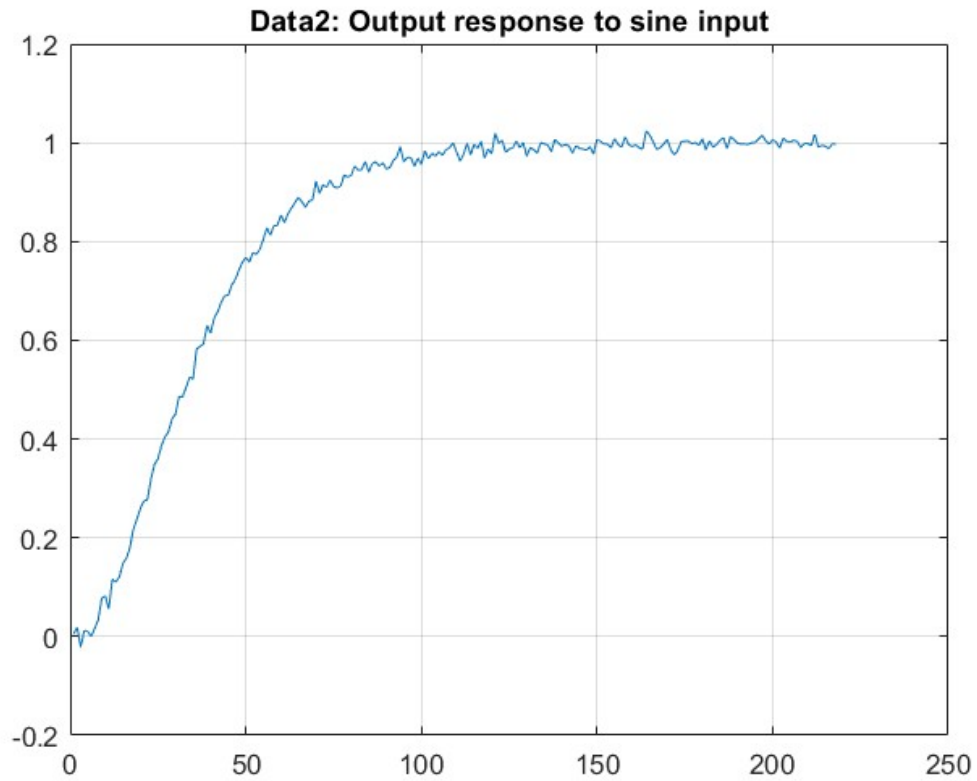


Figure 4: System response corresponding to sine excitation (Data2).

```

Command Window

>> tfest(data2 , 1 , 0 )

ans =

    From input "u1" to output "y1":
    0.2044
    -----
    s + 5.068e-09

Continuous-time identified transfer function.

Parameterization:
    Number of poles: 1   Number of zeros: 0
    Number of free coefficients: 2
    Use "tfdata", "getpvec", "getcov" for parameters and their uncertainties.

Status:
    Estimated using TFEST on time domain data "Data2 (sine excitation)".
    Fit to estimation data: -0.8137%
    FPE: 0.08546, MSE: 0.08314

Model Properties

```

Figure 5: First-order transfer function estimated from Data2 using tfest.

(d) White-noise excitation dataset (Data3)

To provide broadband excitation and assess robustness of the identified model, the input signal was replaced by zero-mean Gaussian white noise, again preserving the sampling time and signal length. The resulting output response constitutes Data3. Due to its wide frequency content, white-noise excitation is particularly suitable for validating low-order dynamic models.

Applying `ttest` to Data3 yields

$$\hat{G}_3(s) = \frac{\hat{k}_3}{s + \hat{a}_3}. \quad (4)$$

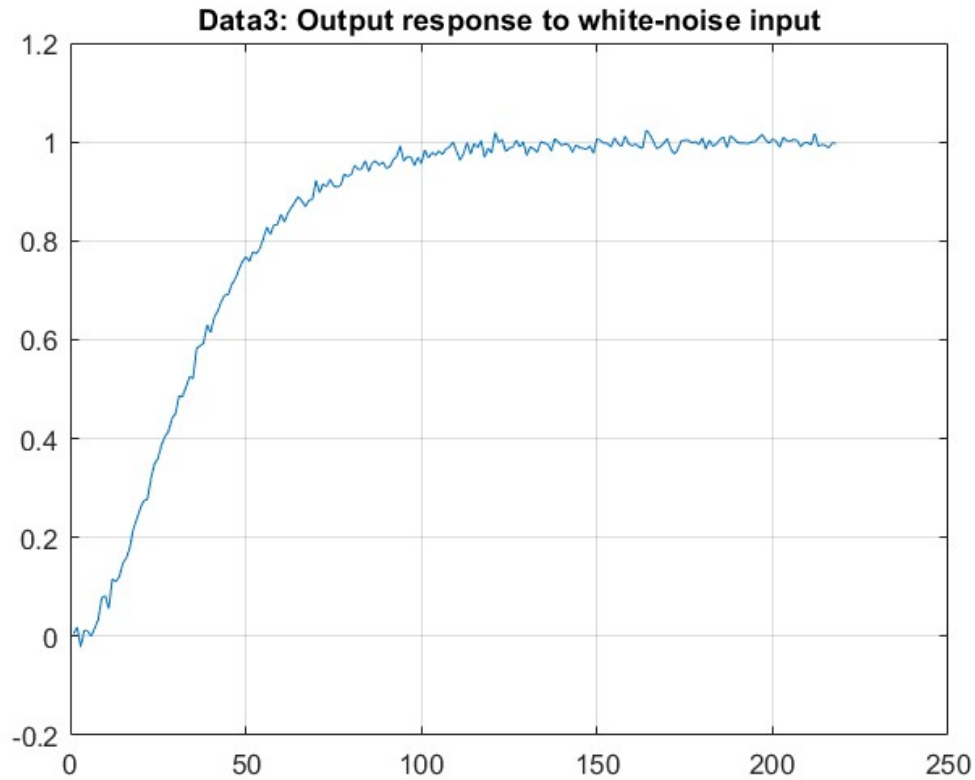


Figure 6: System response corresponding to white-noise excitation (Data3).

```

>> tfest(data3 , 1 , 0 )

ans =

    From input "u1" to output "y1":
    -0.5481
    -----
    s + 3.227e-10

Continuous-time identified transfer function.

Parameterization:
    Number of poles: 1    Number of zeros: 0
    Number of free coefficients: 2
    Use "tfdata", "getpvec", "getcov" for parameters and their uncertainties.

Status:
    Estimated using TFEST on time domain data "Data3 (white-noise excitation)".
    Fit to estimation data: 14.04%
    FPE: 0.06213, MSE: 0.06045

```

Figure 7: First-order transfer function estimated from Data3 using tfest.

Discussion

The identified models obtained from Data1, Data2, and Data3 exhibit comparable pole locations, indicating that the dominant system dynamics are consistently captured by a first-order approximation. Variations in the estimated gain and pole parameters can be attributed to differences in excitation characteristics, with white-noise input generally providing more informative broadband excitation than sinusoidal input.

(e) Low-pass filtering of Data3 and system identification

The dataset Data3, obtained using white-noise excitation, contains significant high-frequency components that can adversely affect parameter estimation. To mitigate this effect, a first-order low-pass filter of the form

$$F(s) = \frac{1}{a_f s + 1} \quad (5)$$

was applied to both the input and output signals of Data3. Filtering both signals ensures that the relative phase and gain between the input and output are preserved.

The filtered signals were then used to construct a new identification dataset, denoted as Data3_f. Using this filtered dataset, a first-order continuous-time transfer function model was estimated with the System Identification Toolbox. The resulting model is given by

$$\hat{G}_{3,f}(s) = \frac{\hat{k}_{3,f}}{s + \hat{a}_{3,f}}. \quad (6)$$

The identified transfer function obtained after noise reduction is shown in Fig. 8. Compared to the unfiltered case, the filtered-data model exhibits smoother dynamics and more stable parameter estimates, indicating that the effect of measurement noise has been successfully reduced.

```

Command Window

Identified model from filtered Data3:

G3_f =

    From input "u1" to output "y1":
        -0.7564
        -----
        s + 3.252e-09

Continuous-time identified transfer function.

Parameterization:
    Number of poles: 1    Number of zeros: 0
    Number of free coefficients: 2
    Use "tfdata", "getpvec", "getcov" for parameters and their uncertainties.

Status:
    Estimated using TFEST on time domain data "Data3 (low-pass filtered)".
    Fit to estimation data: 18.94%
    FPE: 0.06298, MSE: 0.06127

```

Figure 8: First-order transfer function $\hat{G}_{3,f}(s)$ identified from low-pass filtered Data3.

(f) Model comparison and discussion

In this study, four first-order models were identified using different datasets and preprocessing strategies:

- $\hat{G}_1(s)$: identified from measured data (Data1),
- $\hat{G}_2(s)$: identified from sine-excited data (Data2),
- $\hat{G}_3(s)$: identified from white-noise-excited data (Data3),
- $\hat{G}_{3,f}(s)$: identified from filtered white-noise data (Data3_f).

The pole parameter a determines the dominant time constant of the system, $\tau = 1/a$, and therefore plays a critical role in capturing the system dynamics. Across the unfiltered datasets, the estimated pole locations are generally consistent, indicating that a first-order model is sufficient to represent the dominant behavior of the system. However, the estimates obtained from Data3 exhibit greater variability due to the broadband nature of the white-noise excitation and the presence of measurement noise.

After applying the low-pass filter, the model $\hat{G}_{3,f}(s)$ demonstrates improved parameter stability compared to $\hat{G}_3(s)$. This improvement confirms that high-frequency noise can significantly degrade identification accuracy when broadband excitation is used without preprocessing. By attenuating these noise components, the filtered dataset enables more reliable estimation of both the system gain and pole location.

Overall, the comparison shows that:

- Sine excitation (Data2) provides consistent but narrowband identification results.
- White-noise excitation (Data3) excites a wide frequency range but is more sensitive to noise.
- Low-pass filtering of Data3 yields the most reliable balance between excitation richness and noise suppression.

Based on these observations, the filtered white-noise–based model $\hat{G}_{3,f}(s)$ is considered the most reliable first-order approximation of the system dynamics among the models studied.